

UNIT # 01: CELL AND SUB CELLULAR STRUCTURES

(150 MCQs 100 Straight forward + 50 Scenario based)

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
1	Which feature most directly distinguishes a eukaryotic cell from a prokaryotic cell?	Presence of ribosomes	Presence of cytoplasm	Membrane-bound nucleus	Presence of plasma membrane	☐ ☐ ☐ ☐
2	If the plasma membrane loses its selective permeability, which cellular property would be affected first?	Regulation of intracellular composition	DNA replication	Ribosomal assembly	Chromosome condensation	☐ ☐ ☐ ☐
3	The fluid mosaic model describes the plasma membrane primarily as a structure containing	A rigid lipid layer with fixed proteins	A phospholipid bilayer with mobile proteins	A carbohydrate bilayer surrounding proteins	A protein bilayer containing immobile lipids	☐ ☐ ☐ ☐
4	Which membrane component contributes most directly to recognition between neighboring cells?	Cholesterol	Glycoproteins	Phospholipid tails	Cytosolic enzymes	☐ ☐ ☐ ☐
5	A cell placed in a hypotonic solution will tend to gain water because	Solute concentration is higher outside	Water potential favors inward movement of water	ATP drives water into the cell	Proteins actively pump water inward	☐ ☐ ☐ ☐
6	Which organelle is directly associated with synthesis of proteins destined for secretion?	Smooth ER	Rough ER	Lysosome	Peroxisome	☐ ☐ ☐ ☐
7	Ribosomes attached to rough ER differ functionally from free ribosomes mainly in the	Type of ribosomal RNA	Location of protein synthesis products	Mechanism of peptide bond formation	Genetic information they contain	☐ ☐ ☐ ☐
8	The Golgi complex is particularly important for	DNA replication	Modification and sorting of proteins	ATP synthesis	Fatty-acid oxidation only	☐ ☐ ☐ ☐
9	A vesicle carrying a newly synthesized secretory protein normally buds from the	Lysosome	Golgi apparatus	Nucleus	Mitochondrial matrix	☐ ☐ ☐ ☐
10	Which organelle contains hydrolytic enzymes capable of digesting cellular material?	Peroxisome	Lysosome	Ribosome	Centriole	☐ ☐ ☐ ☐
11	The acidic interior of a lysosome is important because it	Prevents enzyme activity	Provides optimum conditions for many hydrolases	Produces ATP	Synthesizes membrane proteins	☐ ☐ ☐ ☐
12	Peroxisomes are especially associated with reactions involving	DNA transcription	Hydrogen peroxide metabolism	Ribosomal synthesis	Protein translation	☐ ☐ ☐ ☐
13	Catalase in peroxisomes protects cells primarily by converting	Oxygen into glucose	Hydrogen peroxide into less harmful products	Glucose into glycogen	ATP into ADP	☐ ☐ ☐ ☐
14	Glyoxisomes are especially important in germinating oil-rich seeds because they facilitate conversion of	Proteins into nucleic acids	Stored lipids toward carbohydrates	Cellulose into proteins	DNA into RNA	☐ ☐ ☐ ☐
15	The major function of the mitochondrial inner membrane is associated with	DNA transcription only	Electron transport and ATP production	Ribosome destruction	Lysosomal digestion	☐ ☐ ☐ ☐

16	Cristae increase mitochondrial efficiency primarily by	Increasing matrix volume	Increasing inner membrane surface area	Increasing DNA quantity	Decreasing enzyme concentration	☐ ☐ ☐ ☐
17	The mitochondrial matrix contains enzymes primarily involved in	Glycolysis	Krebs cycle reactions	DNA translation in nucleus	Golgi modification	☐ ☐ ☐ ☐
18	Which evidence supports the endosymbiotic origin of mitochondria?	They contain cellulose	They contain their own DNA and ribosomes	They lack membranes	They occur only in prokaryotes	☐ ☐ ☐ ☐
19	Chlorophyll molecules responsible for capturing light energy are located mainly in the	Chloroplast envelope	Thylakoid membranes	Stroma only	Outer nuclear membrane	☐ ☐ ☐ ☐
20	Stacks of thylakoids within chloroplasts are called	Cristae	Grana	Cisternae	Matrix	☐ ☐ ☐ ☐
21	The Calvin cycle occurs primarily in the	Thylakoid lumen	Stroma	Outer membrane	Intermembrane space	☐ ☐ ☐ ☐
22	The immediate role of chloroplast electron transport is to help establish	A DNA gradient	A proton gradient	A glucose gradient	A calcium gradient only	☐ ☐ ☐ ☐
23	Which plastid is specialized primarily for storage of starch?	Amyloplast	Chromoplast	Chloroplast	Elaioplast	☐ ☐ ☐ ☐
24	Chromoplasts are primarily associated with	Protein synthesis	Pigment accumulation other than chlorophyll	ATP synthesis	DNA replication	☐ ☐ ☐ ☐
25	The plant cell wall provides structural support mainly through a matrix containing	Cellulose and other polysaccharides	Phospholipids only	DNA and RNA	Cholesterol only	☐ ☐ ☐ ☐
26	Plasmodesmata are important because they	Produce ATP	Connect the cytoplasm of adjacent plant cells	Digest proteins	Replicate DNA	☐ ☐ ☐ ☐
27	Which structure is absent from a typical prokaryotic cell?	Plasma membrane	Ribosomes	Membrane-bound nucleus	Cytoplasm	☐ ☐ ☐ ☐
28	Prokaryotic DNA is generally located in the	Nucleolus	Nucleoid region	Golgi complex	Lysosome	☐ ☐ ☐ ☐
29	Which statement about prokaryotic ribosomes is correct?	They are absent	They are smaller than typical eukaryotic cytoplasmic ribosomes	They occur only inside nuclei	They contain membrane-bound compartments	☐ ☐ ☐ ☐
30	The nucleolus is primarily involved in	Lipid digestion	Ribosomal subunit production	ATP synthesis	Endocytosis	☐ ☐ ☐ ☐
31	Nuclear pores regulate	Movement between nucleus and cytoplasm	Movement through lysosomes	Protein folding in mitochondria	Cell-wall synthesis	☐ ☐ ☐ ☐
32	Chromatin consists mainly of	DNA associated with proteins	RNA associated with lipids	Lipids associated with cellulose	Proteins associated with ATP	☐ ☐ ☐ ☐
33	Which organelle is directly continuous with the outer nuclear membrane?	Rough ER	Lysosome	Golgi complex	Peroxisome	☐ ☐ ☐ ☐
34	Smooth ER is particularly important in cells specialized for	Lipid synthesis and detoxification	Ribosomal RNA production	DNA packaging	Cell division only	☐ ☐ ☐ ☐
35	The Golgi apparatus can be considered a cellular processing center because it	Replicates chromosomes	Modifies, sorts, and packages molecules	Produces all ATP	Performs photosynthesis	☐ ☐ ☐ ☐
36	Secretory vesicles release their	Endocytosis	Exocytosis	Phagocytosis	Osmosis	☐ ☐ ☐ ☐

	contents outside a cell through					
37	Endocytosis requires the plasma membrane to	Become permanently rigid	Invaginate and form vesicles	Dissolve completely	Lose all proteins	☐ ☐ ☐ ☐
38	Phagocytosis is best described as uptake of	Large particles or cells	Water molecules only	Dissolved gases	Small ions through channels	☐ ☐ ☐ ☐
39	Pinocytosis primarily involves uptake of	Extracellular fluid and dissolved substances	Whole cells only	Chromosomes	Mitochondria	☐ ☐ ☐ ☐
40	Receptor-mediated endocytosis is more selective than ordinary pinocytosis because it	Requires no membrane	Depends on specific receptor-ligand interactions	Occurs only in plants	Does not form vesicles	☐ ☐ ☐ ☐
41	Cell signalling allows a cell to	Ignore extracellular information	Detect and respond to environmental or cellular signals	Eliminate its nucleus	Stop all metabolism	☐ ☐ ☐ ☐
42	A signalling molecule that binds to a receptor on the plasma membrane is most likely	Unable to enter the cell directly	Always a steroid	Always an ion	Always a carbohydrate	☐ ☐ ☐ ☐
43	Steroid hormones commonly act through intracellular receptors because they	Are lipid-soluble	Are large proteins	Cannot cross membranes	Are permanently charged	☐ ☐ ☐ ☐
44	A membrane receptor is most directly involved in converting an extracellular signal into	A change in intracellular activity	A chromosome	A cell wall	A ribosome	☐ ☐ ☐ ☐
45	Signal transduction commonly involves	Receptors, relay molecules, and intracellular effectors	Only DNA molecules	Only lysosomal enzymes	Only membrane lipids	☐ ☐ ☐ ☐
46	Which is an example of a second messenger?	cAMP	DNA	Cellulose	Hemoglobin	☐ ☐ ☐ ☐
47	Protein kinases regulate cellular processes primarily by	Adding phosphate groups to target proteins	Removing DNA bases	Destroying ribosomes	Synthesizing cellulose	☐ ☐ ☐ ☐
48	Phosphatases generally oppose kinase activity by	Adding phosphate groups	Removing phosphate groups	Producing ATP from glucose	Producing DNA	☐ ☐ ☐ ☐
49	Signal amplification means that	One signal can produce many downstream activated molecules	Signals become physically larger	Receptors become DNA	The cell stops responding	☐ ☐ ☐ ☐
50	Which sequence best represents a generalized cell-signalling pathway?	Response → receptor → signal → relay	Signal → receptor → transduction → response	DNA → lysosome → signal → receptor	Ribosome → signal → nucleus → membrane	☐ ☐ ☐ ☐
51	A receptor's specificity depends primarily on	Complementarity between receptor and signalling molecule	Cell size	Number of mitochondria	Amount of cytoplasm	☐ ☐ ☐ ☐
52	Why can the same extracellular signal produce different responses in different cells?	Cells may possess different receptors or intracellular signalling machinery	All cells contain identical proteins	Signals change their chemical identity randomly	DNA is absent in some cells	☐ ☐ ☐ ☐
53	The cytoplasm of a eukaryotic cell includes	Cytosol and suspended organelles outside the nucleus	Only the nucleus	Only the plasma membrane	Only extracellular fluid	☐ ☐ ☐ ☐

54	Cytosol is best described as	The aqueous component of the cytoplasm	The entire cell wall	The nuclear membrane	The mitochondrial DNA	☐ ☐ ☐ ☐
55	Which organelle is not surrounded by a membrane?	Lysosome	Ribosome	Peroxisome	Golgi complex	☐ ☐ ☐ ☐
56	Centrioles are most directly associated with	Organization of microtubules during cell division	Protein digestion	Photosynthesis	Lipid storage	☐ ☐ ☐ ☐
57	A typical centrosome contains	A pair of centrioles and associated material	A pair of lysosomes	Chloroplasts	Golgi cisternae	☐ ☐ ☐ ☐
58	Which structure is most directly responsible for maintaining chromosome separation during mitosis?	Spindle microtubules	Lysosomal membrane	Cell wall	Golgi vesicles	☐ ☐ ☐ ☐
59	Vacuoles in plant cells contribute significantly to	Storage and maintenance of cell turgor	Ribosome production	DNA replication	Oxidative phosphorylation	☐ ☐ ☐ ☐
60	The large central vacuole of a plant cell is bounded by the	Tonoplast	Nuclear envelope	Cell wall	Plasmalemma only	☐ ☐ ☐ ☐
61	Which organelle is especially prominent in cells specialized for secretion?	Rough ER and Golgi complex	Centrioles only	Chloroplasts only	Lysosomes only	☐ ☐ ☐ ☐
62	A protein synthesized on rough ER typically reaches the Golgi apparatus through	Transport vesicles	Nuclear pores	Cristae	Plasmodesmata	☐ ☐ ☐ ☐
63	The cis face of the Golgi apparatus generally	Receives transport vesicles	Releases chromosomes	Produces ATP	Performs glycolysis	☐ ☐ ☐ ☐
64	The trans face of the Golgi apparatus is particularly associated with	Sorting and dispatching processed products	DNA replication	Photosynthesis	Ribosome synthesis	☐ ☐ ☐ ☐
65	Lysosomes can originate functionally from the	Golgi apparatus	Mitochondrial matrix	Nucleolus	Chloroplast stroma	☐ ☐ ☐ ☐
66	Autophagy involves	Lysosomal degradation of cellular components	DNA replication	Cell-wall formation	Photosynthetic carbon fixation	☐ ☐ ☐ ☐
67	Which organelle is particularly involved in detoxifying certain harmful substances?	Smooth ER and peroxisomes	Ribosomes only	Nucleolus	Centrioles	☐ ☐ ☐ ☐
68	Peroxisomes differ from lysosomes because peroxisomes primarily contain enzymes involved in	Oxidative reactions	Hydrolytic digestion only	Ribosome assembly	Protein translation	☐ ☐ ☐ ☐
69	A major advantage of compartmentalization in eukaryotic cells is that it	Allows incompatible reactions to occur in specialized environments	Eliminates the need for enzymes	Prevents all molecular movement	Removes the need for ATP	☐ ☐ ☐ ☐
70	The double membrane surrounding mitochondria is consistent with	Their endosymbiotic evolutionary history	Their role in cell-wall formation	Their origin from lysosomes	Their lack of DNA	☐ ☐ ☐ ☐
71	Mitochondrial DNA is generally	Circular and located within mitochondria	Linear and enclosed in the nucleolus	Absent from mitochondria	Identical to nuclear DNA in organization	☐ ☐ ☐ ☐
72	Which structure contains enzymes responsible for ATP synthesis using the proton gradient?	Inner mitochondrial membrane	Outer mitochondrial membrane	Golgi membrane	Nuclear pore	☐ ☐ ☐ ☐
73	The mitochondrial intermembrane	Electron transport	Glycolysis	Translation	DNA replication	☐ ☐ ☐ ☐

	space becomes enriched in protons during					
74	Chloroplasts and mitochondria are similar because both	Have double membranes and their own genetic material	Perform identical metabolic pathways	Lack ribosomes	Are found in all cells	☐ ☐ ☐ ☐
75	Which chloroplast structure directly contains photosystems?	Thylakoid membrane	Stroma	Outer envelope	Intermembrane space	☐ ☐ ☐ ☐
76	The oxygen released during photosynthesis originates primarily from	Water	Carbon dioxide	Glucose	Chlorophyll	☐ ☐ ☐ ☐
77	ATP and NADPH generated by light-dependent reactions are subsequently used mainly in	Carbon fixation reactions	DNA replication	Lysosomal digestion	Ribosome assembly	☐ ☐ ☐ ☐
78	The cell wall differs from the plasma membrane because the cell wall	Provides rigidity and is not the primary selective barrier	Controls all molecular transport	Contains receptor proteins exclusively	Is made primarily of phospholipids	☐ ☐ ☐ ☐
79	Which structure permits direct cytoplasmic continuity between adjacent plant cells?	Plasmodesmata	Desmosomes	Centrioles	Lysosomes	☐ ☐ ☐ ☐
80	Stem cells are characterized by their capacity for	Self-renewal and differentiation	Photosynthesis only	Permanent specialization	Lysosomal digestion	☐ ☐ ☐ ☐
81	A pluripotent stem cell can generally differentiate into	Many cell types derived from the three germ layers	Only one specialized cell type	Only blood cells	Only reproductive cells	☐ ☐ ☐ ☐
82	A major therapeutic significance of stem cells is their potential to	Replace or repair damaged tissues	Eliminate all mutations	Stop cellular signalling	Prevent every disease	☐ ☐ ☐ ☐
83	Totipotent cells have the greatest developmental potential because they can	Give rise to embryonic and extraembryonic tissues	Form only neurons	Form only blood cells	Produce only muscle cells	☐ ☐ ☐ ☐
84	Which process directly increases plasma-membrane surface area?	Exocytosis	Endocytosis	Lysosomal digestion	Ribosomal translation	☐ ☐ ☐ ☐
85	Which process generally decreases plasma-membrane surface area?	Exocytosis	Endocytosis	Protein synthesis	Photosynthesis	☐ ☐ ☐ ☐
86	If ATP production is severely reduced, which process is most likely to be impaired?	Active vesicular transport	Simple diffusion	Osmosis	Passive facilitated diffusion	☐ ☐ ☐ ☐
87	Membrane proteins can function as	Channels, carriers, receptors, and enzymes	DNA molecules only	Cellulose fibers only	Ribosomal subunits only	☐ ☐ ☐ ☐
88	Cholesterol in an animal-cell membrane helps	Modulate membrane fluidity and stability	Synthesize proteins	Replicate DNA	Digest carbohydrates	☐ ☐ ☐ ☐
89	The hydrophobic region of a phospholipid bilayer is formed mainly by	Fatty-acid tails	Phosphate groups	Carbohydrate chains	Protein heads	☐ ☐ ☐ ☐
90	Integral membrane proteins are characterized by	Association with or penetration into the lipid bilayer	Being completely outside the membrane	Being found only in the nucleus	Being made of cellulose	☐ ☐ ☐ ☐
91	Which organelle would be especially abundant in a cell producing steroid hormones?	Smooth ER	Rough ER only	Lysosomes	Ribosomes only	☐ ☐ ☐ ☐

92	A cell that secretes large amounts of digestive enzymes would be expected to contain abundant	Rough ER, Golgi complex, and secretory vesicles	Chloroplasts and glyoxisomes	Centrioles and vacuoles only	Smooth ER only	☐ ☐ ☐ ☐
93	The primary function of ribosomes is	Translation of mRNA into polypeptides	Replication of DNA	Lipid oxidation	ATP storage	☐ ☐ ☐ ☐
94	Which sequence correctly represents the flow of genetic information during protein synthesis?	DNA → RNA → protein	RNA → DNA → lipid	Protein → DNA → RNA	DNA → protein → RNA	☐ ☐ ☐ ☐
95	The nuclear envelope is structurally related to the	Endoplasmic reticulum	Lysosome	Chloroplast	Centriole	☐ ☐ ☐ ☐
96	Nuclear localization of genetic material provides an advantage because it	Separates transcriptional processes from cytoplasmic translation	Prevents RNA synthesis	Eliminates ribosomes	Stops gene regulation	☐ ☐ ☐ ☐
97	Which combination is characteristic of eukaryotic cells but not typical prokaryotic cells?	Nucleus, mitochondria, Golgi apparatus	Ribosomes, DNA, plasma membrane	Cytoplasm, DNA, ribosomes	Plasma membrane, cytoplasm, genetic material	☐ ☐ ☐ ☐
98	Which structure is common to both prokaryotic and eukaryotic cells?	Plasma membrane	Golgi complex	Mitochondrion	Membrane-bound nucleus	☐ ☐ ☐ ☐
99	If lysosomal membranes become damaged extensively, the most immediate consequence could be	Release of hydrolytic enzymes into the cytoplasm	Increased photosynthesis	Enhanced DNA replication	Increased ribosome synthesis	☐ ☐ ☐ ☐
100	A cell's ability to coordinate an extracellular stimulus with a specific intracellular response depends most directly on	Signal transduction pathways	Cell-wall thickness	Number of chromosomes alone	Vacuole size	☐ ☐ ☐ ☐

SCENARIO-BASED MCQs (101–150)

Scenario 1

A researcher studies a secretory pancreatic cell. The cell contains extensive rough ER, numerous Golgi stacks, and many membrane-bound vesicles. A drug is then introduced that prevents vesicles from fusing with the plasma membrane.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
101	Which cellular process would be most directly inhibited by the drug?	Exocytosis	Endocytosis	Diffusion	Osmosis	☐ ☐ ☐ ☐
102	The proteins released by this cell were most likely initially synthesized on	Ribosomes attached to rough ER	Free ribosomes in cytosol only	Mitochondrial ribosomes only	Lysosomal membranes	☐ ☐ ☐ ☐
103	Before reaching the plasma membrane, these proteins would most likely be modified and sorted in the	Golgi complex	Nucleolus	Peroxisome	Centriole	☐ ☐ ☐ ☐
104	If Golgi function were disrupted instead, which event would be affected most directly?	Modification and sorting of secretory proteins	DNA replication	Glycolysis	Formation of spindle fibers	☐ ☐ ☐ ☐

Scenario 2

BIOLOGY HSSC - I

An animal cell is exposed to a signalling molecule that cannot freely cross the phospholipid bilayer. The molecule binds to a receptor on the plasma membrane and activates an intracellular pathway involving cAMP and protein kinases.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
105	The signalling molecule most likely acts through	A membrane receptor	A mitochondrial DNA sequence	A ribosomal binding site	A lysosomal enzyme	☐ ☐ ☐ ☐
106	cAMP in this pathway functions primarily as a	Second messenger	First messenger	Structural lipid	Transcription factor	☐ ☐ ☐ ☐
107	Activation of protein kinases would most directly result in	Phosphorylation of target proteins	Removal of all membrane proteins	DNA degradation	Formation of cellulose	☐ ☐ ☐ ☐
108	If the receptor were absent, the most likely consequence would be	Failure to initiate the specific signalling pathway	Increased ATP synthesis automatically	Increased lysosomal digestion	Immediate DNA replication	☐ ☐ ☐ ☐

Scenario 3

A plant cell is placed under intense illumination. Microscopic analysis shows numerous chloroplasts with extensive stacks of thylakoids. A mutation prevents proper development of thylakoid membranes.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
109	Which photosynthetic process would be most directly impaired?	Light-dependent reactions	Calvin cycle enzymes in the stroma	Cytoplasmic glycolysis	Protein digestion	☐ ☐ ☐ ☐
110	The photosystems are normally located in the	Thylakoid membranes	Chloroplast stroma	Outer chloroplast membrane	Cell wall	☐ ☐ ☐ ☐
111	A reduction in electron transport would most directly reduce production of	ATP and NADPH	DNA and RNA	Lysosomal enzymes	Cellulose and lignin	☐ ☐ ☐ ☐
112	The Calvin cycle would be affected indirectly because it depends on	ATP and NADPH generated by light-dependent reactions	Lysosomal enzymes	Ribosomes in the nucleus	Mitochondrial DNA only	☐ ☐ ☐ ☐

Scenario 4

A scientist examines two cells. Cell X has a circular DNA molecule in a nucleoid region and lacks membrane-bound organelles. Cell Y contains a nucleus, mitochondria, ER, and Golgi apparatus.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
113	Cell X is most likely	Prokaryotic	Eukaryotic	Multinucleate	A mature plant cell	☐ ☐ ☐ ☐
114	Which structure should be present in both cells?	Ribosomes	Golgi apparatus	Nuclear envelope	Mitochondria	☐ ☐ ☐ ☐
115	Cell Y can compartmentalize metabolic reactions more extensively because it possesses	Membrane-bound organelles	Circular DNA only	Smaller ribosomes	A nucleoid	☐ ☐ ☐ ☐
116	If the nuclear envelope of Cell Y disappeared permanently, which distinction between the two cells would become most directly affected?	Separation of genetic material from cytoplasm	Presence of ribosomes	Presence of plasma membrane	Presence of DNA	☐ ☐ ☐ ☐

Scenario 5

A cell is exposed to a chemical that prevents formation of ATP. The cell must continue maintaining ion gradients and internal organization.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
117	Which process would be most directly affected?	ATP-dependent active transport	Simple diffusion	Osmosis	Passive facilitated diffusion	☐ ☐ ☐ ☐
118	Vesicle trafficking would likely be impaired because	Many steps require cellular energy	Vesicles cannot contain proteins	Membranes become DNA	Golgi complexes disappear immediately	☐ ☐ ☐ ☐
119	Which process could continue without direct ATP expenditure?	Simple diffusion down a concentration gradient	Primary active transport	Vesicle formation requiring energy	Protein phosphorylation by ATP-dependent kinases	☐ ☐ ☐ ☐
120	Severe ATP depletion would particularly compromise maintenance of	Electrochemical gradients	DNA base pairing	Phospholipid structure itself	Ribosomal RNA sequence	☐ ☐ ☐ ☐

Scenario 6

A germinating oil-rich seed is examined. Its cells contain numerous specialized microbodies that contain enzymes allowing stored fatty acids to contribute to carbohydrate production.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
121	The specialized microbodies described are	Glyoxisomes	Lysosomes	Ribosomes	Centrioles	☐ ☐ ☐ ☐
122	Their activity is especially important because the seed	Contains stored lipids that can be converted toward carbohydrate metabolism	Cannot perform respiration	Lacks mitochondria	Cannot synthesize enzymes	☐ ☐ ☐ ☐
123	Glyoxisomes are developmentally/functionally related to	Peroxisomes	Golgi complexes	Nuclei	Chloroplast grana	☐ ☐ ☐ ☐
124	The metabolic advantage is greatest during early growth because	Stored reserves can support seedling development before extensive photosynthesis	Photosynthesis is already maximal	The seed has no DNA	Ribosomes cannot function	☐ ☐ ☐ ☐

Scenario 7

A patient has cells in which catalase activity is severely reduced. Laboratory analysis indicates accumulation of hydrogen peroxide in cellular compartments containing oxidative enzymes.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
125	The affected organelles are most likely	Peroxisomes	Lysosomes	Ribosomes	Golgi bodies	☐ ☐ ☐ ☐
126	The accumulated substance is potentially harmful because hydrogen peroxide	Can participate in damaging oxidative reactions	Is a structural component of DNA	Produces ribosomes	Is required for chromosome separation	☐ ☐ ☐ ☐
127	Catalase normally helps protect cells by	Converting hydrogen peroxide into less harmful products	Synthesizing ATP directly	Producing glucose	Digesting proteins	☐ ☐ ☐ ☐
128	Which organelle would be less directly involved in this defect?	Golgi apparatus	Peroxisome	Mitochondrion	Cytosolic enzymes	☐ ☐ ☐ ☐

Scenario 8

A plant cell is placed in a solution with lower solute concentration than the cell sap. After several minutes, the cell becomes more turgid.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
129	The primary movement responsible for the change is	Water entering the cell by osmosis	Solute leaving by exocytosis	Proteins entering by endocytosis	ATP entering through channels	☐ ☐ ☐ ☐
130	The structure that resists excessive expansion is primarily the	Cell wall	Ribosome	Golgi apparatus	Nucleolus	☐ ☐ ☐ ☐
131	The large vacuole contributes to turgidity mainly by	Accumulating water and solutes	Synthesizing DNA	Producing ribosomes	Performing translation	☐ ☐ ☐ ☐
132	The membrane surrounding the vacuole is the	Tonoplast	Nuclear envelope	Crista	Thylakoid	☐ ☐ ☐ ☐

Scenario 9

A drug prevents lysosomal enzymes from functioning. Cellular debris and worn-out organelles begin accumulating inside the cell.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
133	The accumulating material would normally be degraded primarily by	Lysosomal hydrolytic enzymes	Ribosomes	Centrioles	Chlorophyll	☐ ☐ ☐ ☐
134	The recycling of damaged organelles through lysosomes is called	Autophagy	Exocytosis	Translation	Photolysis	☐ ☐ ☐ ☐
135	Lysosomal enzymes function optimally in an environment that is generally	Acidic	Strongly alkaline	Completely neutral only	Anaerobic only	☐ ☐ ☐ ☐
136	Which organelle has an important role in supplying many lysosomal enzymes through protein-processing pathways?	Golgi complex	Mitochondrion	Chloroplast	Centriole	☐ ☐ ☐ ☐

Scenario 10

A cell is engineered to produce a membrane receptor. The receptor is synthesized inside the cell, processed, transported in vesicles, and eventually inserted into the plasma membrane.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
137	The receptor protein would most likely be synthesized on	Ribosomes associated with rough ER	Free DNA molecules	Lysosomal ribosomes	Centrioles	☐ ☐ ☐ ☐
138	The next major processing station is the	Golgi complex	Nucleolus	Mitochondrial matrix	Vacuole	☐ ☐ ☐ ☐
139	Transport between these compartments is mediated largely by	Vesicles	Chromosomes	Plasmodesmata	Cristae	☐ ☐ ☐ ☐
140	Insertion of the receptor into the plasma membrane most directly involves	Membrane fusion during exocytosis	Lysosomal digestion	DNA replication	Phagocytosis	☐ ☐ ☐ ☐

Scenario 11

A researcher adds a signalling molecule to two different cell types. Both cells possess receptors capable of binding the molecule, but Cell A increases enzyme activity while Cell B alters gene expression.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
141	The different responses can best be explained by differences in	Intracellular signalling machinery	Genetic code itself	Plasma membrane phospholipids only	Number of chromosomes only	☐ ☐ ☐ ☐
142	The initial common event is most likely	Ligand-receptor binding	DNA replication	Protein degradation	ATP hydrolysis in lysosomes	☐ ☐ ☐ ☐
143	If Cell A possesses a kinase absent in Cell B, the difference would most likely occur during	Signal transduction	Ligand synthesis outside the cell	Cell-wall formation	Ribosome assembly	☐ ☐ ☐ ☐
144	The observation demonstrates that cellular signalling is	Context-dependent	Identical in every cell	Independent of receptors	Restricted to prokaryotes	☐ ☐ ☐ ☐

Scenario 12

A microscope image shows a cell containing paired cylindrical structures near the nucleus. During cell division, a network of microtubules extends from the surrounding region.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
145	The paired cylindrical structures are	Centrioles	Lysosomes	Peroxisomes	Ribosomes	☐ ☐ ☐ ☐
146	Their associated region contributes to organization of	Microtubules	Cellulose	Thylakoids	DNA bases	☐ ☐ ☐ ☐
147	The microtubule network is especially important for	Chromosome movement during cell division	Protein digestion	Photosynthesis	Lipid storage	☐ ☐ ☐ ☐
148	A defect in this organization could most directly disrupt	Accurate chromosome segregation	Lysosomal pH	Chlorophyll synthesis	Cell-wall composition	☐ ☐ ☐ ☐

Scenario 13

A stem-cell laboratory maintains an undifferentiated cell population. Under one set of chemical signals, the cells become neurons; under another set, they become muscle cells.

Sr. No.	Question	Option 1	Option 2	Option 3	Option 4	Bubbles
149	The ability of these cells to produce different specialized cell types demonstrates	Differentiation potential	Permanent specialization	Loss of genetic material	Absence of signalling	☐ ☐ ☐ ☐
150	The chemical signals are important because they can influence	Which genes and cellular pathways become active	The existence of DNA	Whether ribosomes exist	Whether the plasma membrane exists	☐ ☐ ☐ ☐

ANSWER KEY

Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans	Q / Ans
1 / 3	2 / 1	3 / 2	4 / 2	5 / 2	6 / 2	7 / 2	8 / 2	9 / 2	10 / 2
11 / 2	12 / 2	13 / 2	14 / 2	15 / 2	16 / 2	17 / 2	18 / 2	19 / 2	20 / 2
21 / 2	22 / 2	23 / 1	24 / 2	25 / 1	26 / 2	27 / 3	28 / 2	29 / 2	30 / 2

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31 / 1	32 / 1	33 / 1	34 / 1	35 / 2	36 / 2	37 / 2	38 / 1	39 / 1	40 / 2
41 / 2	42 / 1	43 / 1	44 / 1	45 / 1	46 / 1	47 / 1	48 / 2	49 / 1	50 / 2
51 / 1	52 / 1	53 / 1	54 / 1	55 / 2	56 / 1	57 / 1	58 / 1	59 / 1	60 / 1
61 / 1	62 / 1	63 / 1	64 / 1	65 / 1	66 / 1	67 / 1	68 / 1	69 / 1	70 / 1
71 / 1	72 / 1	73 / 1	74 / 1	75 / 1	76 / 1	77 / 1	78 / 1	79 / 1	80 / 1
81 / 1	82 / 1	83 / 1	84 / 1	85 / 2	86 / 1	87 / 1	88 / 1	89 / 1	90 / 1
91 / 1	92 / 1	93 / 1	94 / 1	95 / 1	96 / 1	97 / 1	98 / 1	99 / 1	100 / 1
101 / 1	102 / 1	103 / 1	104 / 1	105 / 1	106 / 1	107 / 1	108 / 1	109 / 1	110 / 1
111 / 1	112 / 1	113 / 1	114 / 1	115 / 1	116 / 1	117 / 1	118 / 1	119 / 1	120 / 1
121 / 1	122 / 1	123 / 1	124 / 1	125 / 1	126 / 1	127 / 1	128 / 1	129 / 1	130 / 1
131 / 1	132 / 1	133 / 1	134 / 1	135 / 1	136 / 1	137 / 1	138 / 1	139 / 1	140 / 1
141 / 1	142 / 1	143 / 1	144 / 1	145 / 1	146 / 1	147 / 1	148 / 1	149 / 1	150 / 1